

DR. GABI WENZEL
Research Scientist · Laboratory Astrophysicist

Massachusetts Institute of Technology
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EDUCATION

Astrophysics, Space and Environmental Sciences, University of Toulouse	Jul 3, 2020
PhD in Laboratory Astrophysics <i>with Dr. Christine Joblin (HDR)</i>	Toulouse, France
Department of Physics, University of Münster	Mar 1, 2017
Master of Science in Physics <i>with Prof. Dr. Helmut Zacharias</i>	Münster, Germany
Department of Physics, University of Münster	Sep 30, 2014
Bachelor of Science in Physics <i>with Prof. Dr. Helmut Zacharias</i>	Münster, Germany

CURRENT POSITIONS

Massachusetts Institute of Technology	Dec 2024 – Present
<i>Research Scientist, Laboratory Astrochemistry</i>	Cambridge, MA, USA

Conducting follow-up laboratory studies on functionalized PAHs to expand on **our recent interstellar discoveries**. Focusing on measuring previously unknown spectra of functionalized PAHs at MIT. Continuing to supervise graduate students while tailoring my own research profile.

Center for Astrophysics Harvard & Smithsonian	Oct 2024 – Present
<i>SAO Visiting Scientist Fellowship</i>	Cambridge, MA, USA

The origin and fate of PAHs at low temperatures remain poorly understood, and our recent detections in dark molecular clouds challenge our understanding of **interstellar PAH chemistry and physics**. In the [McCarthy Group](#), I investigate potential formation and destruction pathways of cosmic PAHs.

PREVIOUS POSITIONS

Massachusetts Institute of Technology	Oct 2022 – Nov 2024
<i>(Senior) Postdoctoral Associate, Laboratory Astrochemistry</i>	Cambridge, MA, USA

Designed and constructed **chirped-pulse Fourier Transform microwave spectrometers** to study molecular systems of astrochemical interest while leading the rotational spectroscopy efforts in the [McGuire Group](#), including supervision of graduate students in the laboratory and beyond.

Center for Interstellar Catalysis (InterCat), Aarhus University	Oct 2020 – Sep 2022
<i>Postdoctoral Research Fellow, Laboratory Astrophysics</i>	Aarhus C, Denmark

Experimentally investigated the **catalytic effects of astrophysically relevant, functionalized PAHs** and the potential formation of the molecular building blocks of life on different cosmic dust grain surface analogues. Led laboratory experiments, data analysis, and dissemination of results, while supervising Bachelor, Master, and PhD students in the astrophysics laboratory.

Institut de Recherche en Astrophysique et Planétologie (IRAP)	Jun 2017 – Jul 2020
<i>Marie Skłodowska-Curie Early Stage Researcher (ESR)</i>	Toulouse, France

Conducted research and received training in the **interdisciplinary field of laboratory astrophysics** within the [MSCA ITN EUROPAH](#) under the supervision of Dr. Christine Joblin. This included collaborations with theoretician Dr. Aude Simon ([LCPQ](#)) and extended stays in the Netherlands working with Dr. Sandra Brünken at the [FELIX Laboratory](#), and in Italy collaborating with Dr. Giacomo Mulas at [INAF, Cagliari](#).

Hidden Analytical Ltd.	Mar 2019 – May 2019
<i>Advanced Systems Analyst</i>	Warrington, United Kingdom

Carried out an internship and gained experience on quadrupole mass spectrometers. Extended surface analysis skills by performing a project combining temperature programmed desorption ([TPD](#)) and secondary

ion mass spectrometry (SIMS) experiments on differently coated silicon samples.

Institute(s) of (Theoretical) Physics, University of Münster

Nov 2013 – Mar 2017

Student Assistant

Münster, Germany

Undergraduate exercise classes and laboratory instruction. See ‘Teaching Experience’ for details.

Technologieförderung Münster GmbH

Feb 2013 – Dec 2014

Reception Administrative Assistant

Münster, Germany

Main responsibilities included customer service, tenant support as well as preparation and follow-up processing of business events.

PEER RECOGNITION

Invited Talks & Seminars	Plenary Speaker at the 249 th AAS Meeting	Salt Lake City, UT, Jan 2027
	Laboratory Astrophysics Workshop II at AAS 249	Salt Lake City, UT, Jan 2027
	Seminar at the Dept. of Chemistry, UNH	New Haven, CT, Oct 2026
	Colloquium at the NRAO and Dept. of Astronomy, UVA	Charlottesville, VA, Oct 2026
	European Conference on Laboratory Astrophysics (ECLA)	Heidelberg, Germany, Sep 2026
	Committee on Space Research (COSPAR) Meeting	Florence, Italy, Aug 2026
	Exploring the Aromatic Universe in the JWST Era	London, Canada, Jul 2026
	LAD Prize Talk at the 248 th AAS Meeting	Pasadena, CA, Jun 2026
	Virtual International Microwave Seminar	Hamburg, Germany, Jul 2025
	Press Briefing at the 246th AAS Meeting	Anchorage, AL, Jun 2025
	3 rd COST NanoSpace Joint Scientific Meeting	Kaunas, Lithuania, May 2025
	Atomic and Molecular Physics Seminar, CfA	Cambridge, MA, Mar 2025
	InterCat Seminar, Aarhus University	Aarhus, Denmark, Mar 2025
	Green Bank Telescope (GBT) Webinar	Green Bank, WV, Dec 2024
	Öberg & Andrews Seminar, CfA	Cambridge, MA, Nov 2024
	Annual German LabAstro Meeting	Tegernsee, Germany, Sep 2024
	CfA Seminar, Center for Astrophysics Harvard & Smithsonian	Cambridge, MA, Mar 2024
	CFEL Molecular and Ultrafast Science Seminars, DESY	Hamburg, Germany, Jan 2023
	Colloquium at the Istituto Nazionale di Astrofisica (INAF)	Cagliari, Italy, Dec 2018
Awards & Grants	2026 LAD Early Career Award	American Astronomical Society, 2026
	Research Support Committee – School of Science	MIT, Cambridge, MA, 2025
	Women in Chemistry+ Travel Grant	MIT, Cambridge, MA, 2025
	Infinite Expansion Award	MIT, Cambridge, MA, 2024
	Best Talk Award	PAHRTEA Meeting Nijmegen, 2019
	LASERLAB-EUROPE Grant	2-month stay at FELIX Laboratory, 2019
	IAU Travel Grant	IAU S350 Laboratory Astrophysics, 2019
	PCMI Travel Grant	Société Française d’Exobiologie, 2018
	WE-Heraeus Travel Grant	DPG Spring Meeting, 2017
	DPG A-Levels Award	Arndt High School, Krefeld, 2010
News & Media	Riesen-Molekül im Weltall aufgespürt , German newspaper	Spektrum, Germany, Jun 2025
	Largest aromatic molecule found in deep space is... , Article	Chemistry World, UK, Jun 2025
	Exploring the Molecular Universe... , Interview	Spectroscopy, USA, May 2025
	Schadstoffe im fast leeren Raum , German newspaper	Die Zeit, Germany, Oct 2024
	Carbon sources spotted in cold interstellar clouds , Article	CE&N, USA, Oct 2024
	Une molécule détectée pour la première fois... , Article	Radio Canada, Oct 2024

SELECTED RESEARCH EXPERIENCE

- Green Bank Telescope (GBO), WV, USA* Mar 2024 – present
Active GOTHAM collaboration observer using the 100 m Green Bank Telescope to study aromatic molecules in Taurus, Serpens, and Horsehead to trace star formation chemistry.
- LISA@FELIX Laboratory, Nijmegen, The Netherlands* May 2022, Sep 2022, Aug 2023
Used IR beamtime to study PAH:ice chemistry at the LISA end-station with FTIR spectroscopy on ASW, CO, and methanol ices.
- ELISA, Aarhus, Denmark* Mar 2022 – Jun 2022
Led ion storage ring experiments using visible photodissociation spectroscopy to study photostability trends of oxygen-functionalized PAHs.
- ASTRID-2 Synchrotron, Aarhus, Denmark* Feb 2021
Performed XPS on oxygen-irradiated C₆₀ on HOPG; observed enhanced chemistry under repeated annealing and exposure.
- CALMIP HPC Center, Toulouse, France* Dec 2019
Ran DFT calculations of PAHs with Turbomole and Octopus; obtained theoretical IR spectra and photoabsorption cross sections.
- SOLEIL Synchrotron, France* Sep 2017, Jul 2019
Used VUV action spectroscopy to study fragmentation and ionization of large PAHs in a linear ion trap.
- FELion@FELIX Laboratory, Nijmegen, The Netherlands* Aug 2017, Jun – Jul 2019, Jan 2023
Modified cryogenic ion trap setup; measured IR action spectra of PAH fragments tagged with Ne.
- DESIREE, Stockholm, Sweden* Nov 2018
Studied radiative cooling of anthracene and phenanthrene cations in an electrostatic storage ring.
- FLASH, DESY, Hamburg, Germany* Aug 2015
Participated in XUV-FEL desorption experiments of astrophysical ices; used TOF-MS to analyze desorption dynamics.

RESEARCH & TECHNICAL SKILLS

Vacuum systems: Ultra-high vacuum (UHV) and high vacuum (HV) equipment use and maintenance
Surface science techniques: AES, LEED, UPS, XPS, IPE, SIMS, STM, TPD, FTIR/RAIRS
Mass spectrometry: Gas-phase ion trap and storage experiments; FT-ICR MS, quadrupole MS (QMS)
Rotational spectroscopy: CP-FTMW, cavity-enhanced FTMW spectroscopy
Photon sources: Use of free electron lasers and synchrotron radiation facilities
Programming and software: Python, LaTeX, C++
Quantum chemistry software: Psi4, Orca, Turbomole, Octopus
Data acquisition and control: LabVIEW, C++/QtCreator
Languages: German (native), English (fluent, C2), French (advanced, B1/B2), Danish (beginner, A2)

TEACHING EXPERIENCE

- Co-Organizer of the Star and Planet Formation Course* Apr 2022 – Oct 2022
Mentoring a group of PhD and Master level students following the star and planet formation lecture given online by Dr. Melissa McClure at Leiden University. Supervised discussion rounds, seminars, short presentations, and essays.
- Tutor for the Undergraduate Modules Physics I – III* Mar 2015 – Mar 2017
Marking of first to second year physics students' exercises and giving class explanations of the solutions. These modules covered mechanics, thermodynamics, electromagnetism, analytical mechanics, electrodynamics, optics, special relativity.

General demonstrating duties assisting undergraduate medical students in their first year performing experimental exercises in the physics laboratory. Responsibilities also included the marking of laboratory reports, students' exercises, and discussions of the solutions during the courses.

LEADERSHIP & OUTREACH

Heading Laboratory Rot. Spec. Efforts
Postdoc Representative
Early Stage Researcher Representative
Active Member of UniverSciEl
Second PhD Student Representative

Dept. of Chemistry, MIT, 2022 – 2026
Dept. of Physics & Astronomy, AU, 2021 – 2022
EUROPAH Network, 2018 – 2021
Science outreach for children, 2017 – 2020
IRAP Toulouse, 2017 – 2019

SCHOLARLY SERVICE & REVIEWING

- Reviewer for: The Astrophysical Journal, Astronomy & Astrophysics, Physical Review Letters, Nature Communications, Journal of Molecular Spectroscopy, Molecular Physics, ACS Earth and Space Chemistry, Physical Chemistry Chemical Physics (2019 – present)
- Expert and panel reviewer, NASA Emerging Worlds Program (2024 and 2025)
- Expert reviewer, ANR (French National Research Agency, 2026)

PUBLICATIONS

[ADS](#) & [Google Scholar](#) 

*25 papers published (9 as first or corresponding author) in international peer-reviewed journals (including Science, Nature Astronomy, The Astrophysical Journal (Letters), Astronomy & Astrophysics, Journal of Physical Chemistry), 3 peer-reviewed conference proceedings, 3 papers in press, 3 papers submitted. Supervisee papers are marked with *[X]. My three most important publications are highlighted in [violet](#) shade.*

*[34] Ren, H.; **Wenzel, G.**; Speak, T. H.; Duffy, M.; Hariharan, A.; Holdren, M. S.; Stewart, D. A.; Shinglededecker, C. N.; Willis, R. H. J.; Cooke, I. R.; McGuire, B. A. Topology-Dependent Selectivity of Interstellar Polycyclic Aromatic Hydrocarbons: Insights from the Absence of Cyanotriphenylenes. Submitted to *The Astrophysical Journal*.

[33] Mehalko, K. T.; Esselman, B. J.; Franke, P. R.; Holdren, M. S.; **Wenzel, G.**; Stewart, D. A.; Burke, Z. P.; McGuire, B. A.; Woods, R. C.; McMahon, R. J. Structure of Furan from 44 Isotopologues – Resolving Differences in the Theoretical (r_e) and Semi-Experimental Equilibrium (r_e^{SE}) Structures. Submitted to *Journal of Physical Chemistry A*.

[32] Esposito, V. J.; Ferrari, P.; Maragkoudakis, A.; **Wenzel, G.**; Smith, B. S.; Gong, S.; Posma, L. J.; Candian, A.; Temi, P.; Bregman, J. D.; Wendlandt, A. E.; McGuire, B. A.; Buma, W. J.; Boersma, C.; Fortenberry, R. C.; Allamandola, L. J.; Tielens, A. G. G. M. Infrared spectroscopy of the CN and CD stretches of substituted polycyclic aromatic hydrocarbons. Submitted to *Nature Communications*.

*[31] Chidester, N.; Stewart, D. A.; Speak, T. H.; Berger, L.; Holdren, M. S.; Hariharan, A.; Lipnicky, A.; Cooke, I. R.; **Wenzel, G.**; McGuire, B. A. On the Astronomical Search and Astrochemical Study of Cyanoisquinolines in TMC-1: Rotational Spectroscopy and Interstellar Investigation. *The Journal of Physical Chemistry*, in press.

[30] Fried, Z. T. P.; Loomis, R. A.; Jørgensen, J. K.; Lipnicky, A.; Xue, C.; **Wenzel, G.**; Speak, T. H.; McCarthy, M. C.; McGuire, B. A. Automated Assignment and Prediction of Molecules in Astronomical Line Surveys Using Machine-Learning-Based Chemical Embeddings. *The Astrophysical Journal*, in press.

[29] Duffy, M.; Loomis, R. A.; Holdren, M. S.; Lipnicky, A.; Stewart, D. A.; **Wenzel, G.**; Xue, C.; Hariharan, A.; Cooke, I. R.; Remijan, A. J.; McCarthy, M. C.; McGuire, B. A. Exploring the Limits of Spectral Line

Stacking in Spectral Line Data and Application Toward the Detection of Bulk ^{13}C Enrichment of Aromatics in TMC-1. *The Astrophysical Journal*, in press.

[28] **Wenzel, G.**; Speak, T. H.; Xue, C.; Bergin, E. A.; Burkhardt, A. M.; Cordiner, M. A.; Duffy, M.; Fried, Z. T. P.; Lipnicky, A.; Shingledecker, C. N.; Willis, R. H. J.; Remijan, A. J.; McCarthy, M. C.; McGuire, B. A.; Cooke, I. R. Detection of the Polycyclic Aromatic Hydrocarbon Phenalene in the Very Low-Luminosity Object MC27/L1521F. *The Astrophysical Journal Letters* 2026, 1006 (2), L38.

*[27] Holdren, M. S.; Fried, Z. T. P.; Stewart, D. A.; **Wenzel, G.**; Speak, T. H.; McGuire, B. A. Rotational Spectra and Interstellar Search for Chiral and Achiral Butynols and Butenols: 3-butyn-2-ol, 3-buten-2-ol, 3-butyn-1-ol, and 3-buten-1-ol. *ACS Earth and Space Chemistry* 2026, 10, 406–420.

[26] Byrne, A. N.; Shingledecker, C. N.; Bergin, E. A.; Holdren, M. S.; **Wenzel, G.**; Xue, C.; Van Voorhis, T.; McGuire, B. A. Modeling the Effect of C/O Ratio on Complex Carbon Chemistry in Cold Molecular Clouds. *The Astrophysical Journal* 2026, 998 (1), 95.

*[25] Ren, H.; Stewart, D. A.; **Wenzel, G.**; Speak, T. H.; Holdren, M. S.; Willis, R. H. J.; McGuire, B. A. Rotational Spectra and Search for Aromatic Imines: 9-Iminofluorene and Benzophenone Imine. *The Astrophysical Journal* 2026, 996 (1), 108.

*[24] Stewart, D. A.; Speak, T. H.; Yuan, E. Q. H.; Willis, R. H. J.; Sconce, V.; Fried, Z. T. P.; Holdren, M. S.; Lipnicky, A.; Shingledecker, C. N.; McCarthy, M. C.; Cooke, I. R.; **Wenzel, G.**; McGuire, B. A. Centimeter-wave Rotational Spectroscopy of Ethynylbenzonitriles: Ground State Structural Parameters and Astronomical Search. *The Journal of Physical Chemistry A* 2025, 129 (49), 11400–11413.

*[23] Slumstrup, L.; Thrower, J. D.; Hopkinson, A. T.; **Wenzel, G.**; Jaganathan, R.; Schrauwen, J. G. M.; Redlich, B.; Ioppolo, S.; Hornekær, L. CO desorption from interstellar icy grains induced by infrared excitation of superhydrogenated polycyclic aromatic hydrocarbons. *Astronomy & Astrophysics* 2025, 704, A74.

*[22] Toru Shay, H.; **Wenzel, G.**; Xue, C.; McGuire, B. A. Bridging Machine Learning and Spectroscopy: A New Analysis for Astrochemical Target Selection. *The Astrophysical Journal* 2025, 993 (1), 75.

[21] Xue, C.; Byrne, A. N.; Morgan, L.; **Wenzel, G.**; Changala, P. B.; Fried, Z. T. P.; Loomis, R. A.; Remijan, A. J.; Bergin, E. A.; Cooke, I. R.; Frayer, D.; Burkhardt, A. M.; Charnley, S. B.; Cordiner, M. A.; Lipnicky, A.; McCarthy, M. C.; McGuire, B. A. The Molecular Inventory of TMC-1 with GOTHAM Observations. *The Astrophysical Journal Supplement Series* 2025, 281 (1), 9.

[20] Slumstrup, L.; Thrower, J. D.; Schrauwen, J. G. M.; Lamberts, T.; Ingman, E. R.; Laurinavicius, D.; DeVine, J.; Terwisscha van Scheltinga, J.; Santos, J. C.; Noble, J. A.; **Wenzel, G.**; McCoustra, M. R. S.; Brown, W. A.; Linnartz, H.; Hornekær, L.; Cuppen, H. M.; Redlich, B.; Ioppolo, S. IR-Induced CO Photodesorption from Pure CO Ice and CO on Amorphous Solid Water. *ACS Earth and Space Chemistry* 2025, 9, 1607–1621.

*[19] Toru Shay, H.; Scolati, H. N.; **Wenzel, G.**; Lee, K. L. K.; Marimuthu, A. N.; McGuire, B. A. Exploring Effects of Modified Machine Learning Pipelines of Astrochemical Inventories. *The Astrophysical Journal* 2025, 985 (1), 123.

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[18] **Wenzel, G.**; Gong, S.; Xue, C.; Changala, P. B.; Holdren, M. S.; Speak, T. H.; Stewart, D. A.; Fried, Z. T. P.; Willis, R. H. J.; Bergin, E. A.; Burkhardt, A. M.; Byrne, A. N.; Charnley, S. B.; Lipnicky, A.; Loomis, R. A.; Shingledecker, C. N.; Cooke, I. R.; McCarthy, M. C.; Remijan, A. J.; Wendlandt, A. E.; McGuire, B. A. Discovery of the Seven-Ring Polycyclic Aromatic Hydrocarbon Cyanocoronene in GOTHAM Obser-

vations of TMC-1. *The Astrophysical Journal Letters* 2025, 984 (1), L36.

[17] **Wenzel, G.**; Holdren, M. S.; Stewart, D. A.; Toru Shay, H.; Byrne, A. N.; Xue, C.; McGuire, B. A. Laboratory Rotational Spectra of Cyanocyclohexane and Its Siblings (1- and 4-Cyanocyclohexene) Using a Compact CP-FTMW Spectrometer for Interstellar Detection. *The Journal of Physical Chemistry A* 2025, 129 (18), 3986–4001.

[16] **Wenzel, G.**; Jiménez-Redondo, M.; Ončák, M.; McGuire, B. A.; Brünken, S.; Caselli, P.; Jusko, P. Infrared Spectroscopy of Pentagon-Containing PAHs: Indenyl and Fluorenyl Anions and Indenyl Cation. *Journal of Physical Chemistry Letters* 2025, 16 (16), 3938–3944.

[15] Remijan, A. J.; Changala, P. B.; Xue, C.; Yuan, E. Q. H.; Duffy, M.; Scolati, H. N.; Shingledecker, C. N.; Speak, T. H.; Cooke, I. R.; Loomis, R.; Burkhardt, A. M.; Fried, Z. T. P.; **Wenzel, G.**; Lipnicky, A.; McCarthy, M. C.; McGuire, B. A. The Missing Link of Sulfur Chemistry in TMC-1: The Detection of $c\text{-C}_3\text{H}_2\text{S}$ from the GOTHAM Survey. *The Astrophysical Journal* 2025, 982 (2), 191.

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[14] **Wenzel, G.**; Speak, T. H.; Changala, P. B.; Willis, R. H. J.; Burkhardt, A. M.; Zhang, S.; Bergin, E. A.; Byrne, A. N.; Charnley, S. B.; Fried, Z. T. P.; Gupta, H.; Herbst, E.; Holdren, M. S.; Lipnicky, A.; Loomis, R. A.; Shingledecker, C. N.; Xue, C.; Remijan, A. J.; Wendlandt, A. E.; McCarthy, M. C.; Cooke, I. R.; McGuire, B. A. Detections of Interstellar Aromatic Nitriles 2-Cyanopyrene and 4-Cyanopyrene in TMC-1. *Nature Astronomy* 2025, 9 (2), 262–270.

[13] Remijan, A. J.; Fried, Z. T. P.; Cooke, I. R.; **Wenzel, G.**; Loomis, R.; Shingledecker, C. N.; Lipnicky, A.; Xue, C.; McCarthy, M. C.; McGuire, B. A. High Spectral Resolution Observations of Propynal (HCCCHO) toward TMC-1 from the GOTHAM Large Program on the GBT. *The Astrophysical Journal* 2024, 976 (1), 105.

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[12] **Wenzel, G.**; Cooke, I. R.; Changala, P. B.; Bergin, E. A.; Zhang, S.; Burkhardt, A. M.; Byrne, A. N.; Charnley, S. B.; Cordiner, M. A.; Duffy, M.; Fried, Z. T. P.; Gupta, H.; Holdren, M. S.; Lipnicky, A.; Loomis, R. A.; Toru Shay, H.; Shingledecker, C. N.; Siebert, M. A.; Stewart, D. A.; Willis, R. H. J.; Xue, C.; Remijan, A. J.; Wendlandt, A. E.; McCarthy, M. C.; McGuire, B. A. Detection of Interstellar 1-Cyanopyrene: A Four-Ring Polycyclic Aromatic Hydrocarbon. *Science* 2024, 386 (6723), 810–813.

*[11] Fried, Z. T. P.; El-Abd, S. J.; Hays, B. M.; **Wenzel, G.**; Byrne, A. N.; Margulès, L.; Motiyenko, R. A.; Shipman, S. T.; Horne, M. P.; Jørgensen, J. K.; Brogan, C. L.; Hunter, T. R.; Remijan, A. J.; Lipnicky, A.; Loomis, R. A.; McGuire, B. A. Rotational Spectrum and First Interstellar Detection of 2-Methoxyethanol Using ALMA Observations of NGC 6334I. *The Astrophysical Journal Letters* 2024, 965 (2), L23.

*[10] Rasmussen, A. P.; **Wenzel, G.**; Hornekær, L.; Andersen, L. H. Gas-Phase Electronic Action Absorption Spectra of Protonated Oxygen-Functionalized Polycyclic Aromatic Hydrocarbons (OPAHs). *Astronomy & Astrophysics* 2023, 674, A103.

[9] **Wenzel, G.**; Simon, A.; Banhatti, S.; Jusko, P.; Schlemmer, S.; Brünken, S.; Joblin, C. Infrared Spectroscopy of the Benzylium-like (and Tropylium-like) Isomers Formed in the $-\text{H}$ Dissociative Ionization of Methylated PAHs. *Journal of Molecular Spectroscopy* 2022, 385, 111620.

- [8] Banhatti, S.; Rap, D. B.; Simon, A.; **Wenzel, G.**; Leboucher, H.; Joblin, C.; Redlich, B.; Schlemmer, S.; Brünken, S. Formation of the Acenaphthylene Cation as a Common C_2H_2 -Loss Fragment in Dissociative Ionization of the PAH Isomers Anthracene and Phenanthrene. *Physical Chemistry Chemical Physics* 2022, 24, 27343–27354.
- [7] **Wenzel, G.**; Joblin, C.; Giuliani, A.; Castillo, S. R.; Mulas, G.; Ji, M.; Sabbah, H.; Quiroga, S.; Peña, D.; Nahon, L. Astrochemical Relevance of VUV Ionization of Large PAH Cations. *Astronomy & Astrophysics* 2020, 641, A98.
- [6] Joblin, C.; **Wenzel, G.**; Castillo, S. R.; Simon, A.; Sabbah, H.; Bonnamy, A.; Toubanc, D.; Mulas, G.; Ji, M.; Giuliani, A.; Nahon, L. Photo-Processing of Astro-PAHs. *Journal of Physics: Conference Series* 2020, 1412, 062002.
- [5] Bernard, J.; Ji, M. C.; Martin, S.; **Wenzel, G.**; Al-Mogeeth, A.; Stockett, M. H.; Schmidt, H. T.; Zettergren, H.; Joblin, C. Radiative Cooling Dynamics of Anthracene Cations Stored in DESIREE Studied via the Time Evolution of 2-Photon-Absorption Induced Dissociation Rate. *Journal of Physics: Conference Series* 2020, 1412, 232013.
- [4] **Wenzel, G.**; Castillo, S. R.; Mulas, G.; Ji, M.-C.; Bonnamy, A.; Sabbah, H.; Giuliani, A.; Nahon, L.; Joblin, C. Photoprocessing of Large PAH Cations. *Proceedings of the International Astronomical Union* 2019, 15 (S350), 388–389.
- [3] Kleimeier, N. F.; **Wenzel, G.**; Urban, A. J.; Tchalala, M. R.; Oughaddou, H.; Dedkov, Y.; Voloshina, E.; Zacharias, H. Unoccupied Electronic Band Structure of Pentagonal Si Nanoribbons on Ag(110). *Physical Chemistry Chemical Physics* 2019, 21 (32), 17811–17820.
- [2] Jusko, P.; Simon, A.; **Wenzel, G.**; Brünken, S.; Schlemmer, S.; Joblin, C. Identification of the Fragment of the 1-Methylpyrene Cation by Mid-IR Spectroscopy. *Chemical Physics Letters* 2018, 698, 206–210.
- [1] Espeter, P.; Keutner, C.; Roesse, P.; Shamout, K.; Berges, U.; **Wenzel, G.**; Bignardi, L.; Kleimeier, N. F.; Zacharias, H.; Westphal, C. Facing the Interaction of Adsorbed Silicon Nano-Ribbons on Silver. *Nanotechnology* 2017, 28 (45), 455701.

CONFERENCE CONTRIBUTIONS

Astrochemistry in the Broadband Era (Poster)	Portland, ME, USA, Oct 2025
International Symposium on Molecular Spectroscopy (2 Talks)	Champaign, IL, USA, Jun 2025
American Astronomical Society, 246 th Meeting (Talk)	Anchorage, AK, USA, Jun 2025
International Symposium on Molecular Spectroscopy (2 Talks)	Champaign, IL, USA, Jun 2024
IAU S383: Astrochemistry VIII (Poster)	Traverse City, MI, USA, Jul 2023
International Symposium on Molecular Spectroscopy (Talk)	Champaign, IL, USA, Jun 2023
Life Cycle of Cosmic PAHs Symposium (Talk & Poster)	Aarhus, Denmark, Sep 2022
739 th WE-Heraeus-Seminar (Talk)	Bad Honnef, Germany, Feb 2022
PAHRTEA Meeting (Talk)	Nijmegen, Netherlands, Sep 2019
Physics and Chemistry of the Interstellar Medium (Poster)	Avignon, France, Sep 2019
IAU S350: Laboratory Astrophysics Symposium (Poster)	Cambridge, UK, Apr 2019
EPoLM-4 Conference (Talk)	Madrid, Spain, Mar 2019
PCMI National Program Symposium (Poster)	Marseille, France, Jun 2018
Cosmic Dust Symposium (Poster)	Copenhagen, Denmark, Jun 2018
DPG Spring Meeting of the Condensed Matter Section (Talk)	Dresden, Germany, Mar 2017



August 11, 2026